Supplementary Material:
Measuring Spatiotemporal Sampling Sensitivity in Video Action Recognition: A Cross-Architecture Analysis at Scale

Anonymous ACCV 2026 Submission

Paper ID #230

This document provides supplementary material for the main paper. We include: (S1) full coverage×stride heatmaps for all 8 models and 8 datasets; (S2) Levene variance inflation analysis; (S3) full ANOVA effect size tables including FineGym; (S4) action sensitivity taxonomy detail; (S5) maximum-probability confidence-cascade curves for all models; (S6) clip duration analysis; (S7) spectral correlation detail; (S8) implementation details (splits, sampling, frame normalization, target-resolution fine-tuning, positional embedding interpolation); (S9) full multi-dataset spatial resolution results (all 8 datasets × 8 models); (S10) deterministic interactive dashboard and code availability; (S11) TDS ranking robustness: leave-one-architecture-out, architecture-family ablation (CNN-only, Transformer-only, Transformer+SSM pools), and bootstrap confidence intervals; (S12) exploratory spectral and label-space evidence-localization checks showing why whole-frame flow is not a literal Nyquist proxy; and (S13) additional robustness checks on optimization headroom, coverage–stride interaction, and held-out cascade calibration.

S1 Full Coverage×Stride Heatmaps

Figures S1–S8 show the complete coverage×stride accuracy heatmaps for all 8 architectures across all 8 datasets. Each cell represents Top-1 accuracy (%) at a specific (coverage, stride) configuration. Color scale: green=high accuracy, red=low accuracy. FineGym heatmaps (all 8 models) are included; per-class optical-flow analysis for FineGym is reported in supplementary S7.

Row order (coverage): 10%, 25%, 50%, 75%, 100%. Column order (stride): 1, 2, 4, 8, 16.

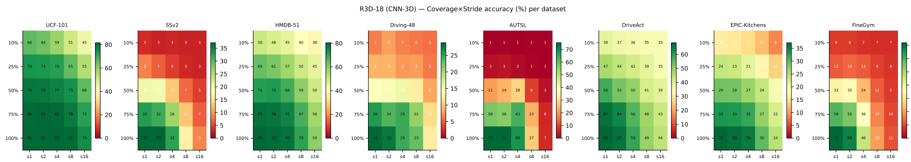


Fig. S1: R3D-18 (CNN-3D) coverage×stride accuracy (%) across 8 datasets. Sharp cliff at stride 4→8 on AUTSL (75%→27%) and similar cliff on FineGym are consistent with the high temporal demand of fine-grained actions. EPIC-Kitchens and UCF-101 retain high accuracy even at sparse sampling.

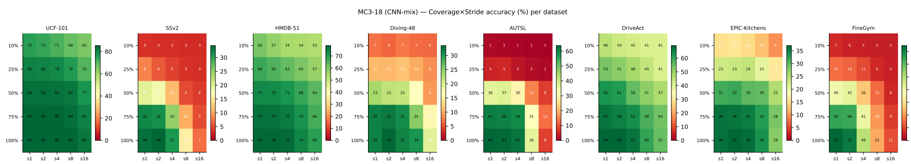


Fig. S2: MC3-18 (CNN-mix, channel-separated 3D) heatmaps. Similar pattern to R3D-18 but with slightly reduced evidence loss on low-TDS datasets (UCF-101, DriveAct). FineGym shows the same severe cliff pattern as AUTSL.

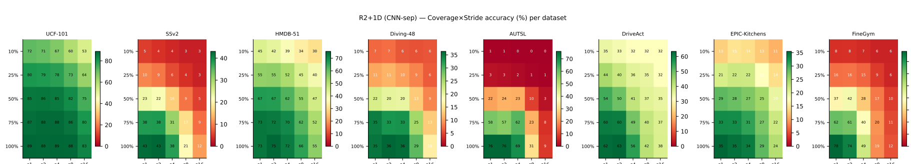


Fig. S3: R2+1D-18 (separable spatiotemporal convolution) heatmaps. Higher base accuracy than R3D-18 on several datasets (SSv2: 42.6% vs 37.1%), but a similar sampling-sensitivity pattern.

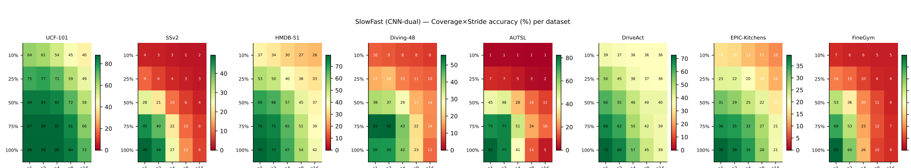


Fig. S4: SlowFast-R50 (dual-pathway) heatmaps. The AUTSL cliff occurs at stride 2→4 (77%→41%), one step earlier than for the other CNNs under the original fixed-budget protocol. This difference is consistent with SlowFast's larger frame budget but does not isolate its causal role. On FineGym, SlowFast achieves the highest base accuracy (78.2%) but suffers the largest evidence loss (71.5pp at stride=16) under this protocol.

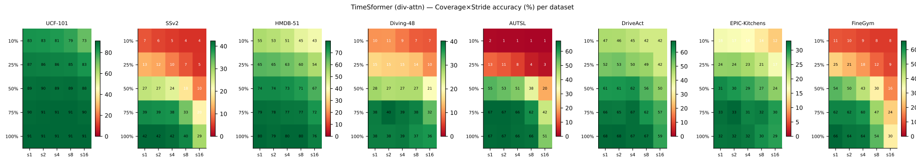


Fig. S5: TimeSformer (divided space–time attention) heatmaps. Accuracy is relatively flat across much of the original grid, with steeper degradation at the combined extreme of low coverage and high stride. FineGym drops 36.1pp at stride=16. These are descriptive fixed-budget curves; they do not identify the contribution of divided attention.

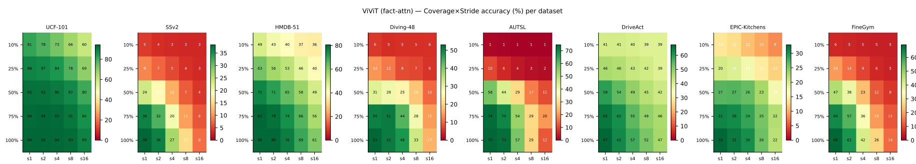


Fig. S6: ViViT (factorized attention) heatmaps. Despite being a Transformer, ViViT shows large fixed-budget drops (AUTSL: cliff at stride 2→4, comparable to SlowFast); FineGym drops 55.2pp at stride=16. The comparison is observational and does not attribute this pattern to factorized temporal attention.

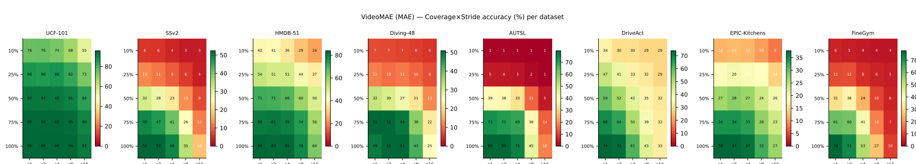


Fig. S7: VideoMAE (masked autoencoder) heatmaps. Base accuracy is high on SSv2 (52.4%), UCF-101 (95.4%), and FineGym (78.0%), while its original-protocol curve falls sharply on AUTSL (stride 4→8) and FineGym (67.8pp at stride=16). This pattern cannot isolate the contribution of the MAE pretraining objective.

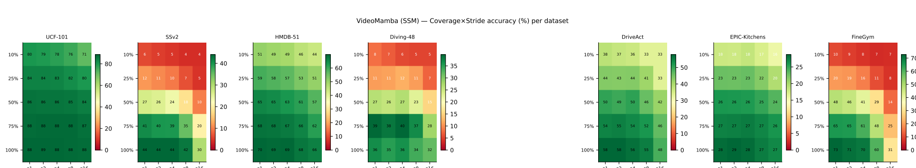


Fig. S8: VideoMamba (SSM selective scan) heatmaps (all 8 datasets). Sensitivity is low on five datasets and higher on FineGym (40.8pp), AUTSL (16.0pp), and SSv2 (14.1pp); these observational differences do not isolate the effect of the temporal module.

S2 Levene Variance Inflation Analysis

Figure S9 plots inter-class standard deviation at stride=1 versus stride=16 for all model-dataset pairs. Points above the diagonal indicate that sparse sampling increases inter-class variability. Significant pairs ($p < 0.05$, Bonferroni-corrected) are highlighted in red.

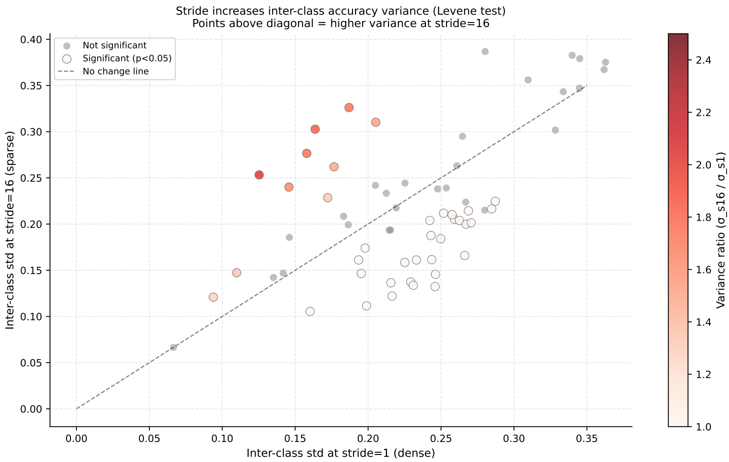


Fig. S9: Levene test: inter-class accuracy std at stride=1 vs. stride=16. Points above the dashed diagonal have higher variance under sparse sampling. Color encodes the variance ratio (σ_{s16}/σ_{s1}). **Key finding**: 67% of model-dataset pairs show significant variance inflation ($p < 0.05$), with the largest increases in VideoMAE/HMDB-51 (2.0×) and R3D-18/HMDB-51 (1.85×). TimeSformer and VideoMamba show stable variance (≈1.1×).

Table S1 lists the 10 model-dataset pairs with highest variance inflation, quantifying the instability masked by mean accuracy.

Table S1: Top-10 variance inflation cases (stride=1→16, coverage=100%). All rows significant at $p<0.05$.

Model	Dataset	$\sigma_{s=1}$	$\sigma_{s=16}$	Ratio	Levene p
VideoMAE	HMDB-51	0.125	0.253	2.02×	<0.0001
R3D-18	HMDB-51	0.164	0.303	1.85×	0.0001
ViViT	HMDB-51	0.158	0.277	1.75×	0.0004
SlowFast	HMDB-51	0.187	0.326	1.74×	<0.0001
SlowFast	UCF-101	0.146	0.240	1.65×	<0.0001
R2+1D	HMDB-51	0.205	0.310	1.51×	0.0021
MC3-18	HMDB-51	0.177	0.262	1.48×	0.0187
ViViT	UCF-101	0.110	0.147	1.34×	0.0093
VideoMAE	AUTSL	0.172	0.229	1.33×	0.0324
VideoMAE	UCF-101	0.094	0.121	1.29×	0.0239

S3 Original Between-Cell ANOVA Effect Sizes

This section preserves the originally submitted between-cell analysis for traceability. Because the same clips populate all 25 cells, its independence assumption is inappropriate; the corrected clip-level repeated-measures analysis in Sec. S16 supersedes it for inference.

Figure S10 shows the mean η^2 per model (averaged over all valid datasets), with error bars indicating standard deviation.

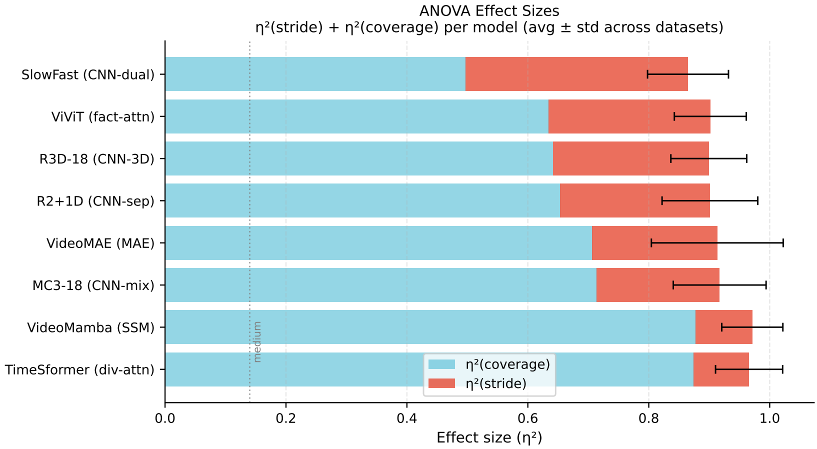


Fig. S10: Originally submitted between-cell η^2 values. These descriptive bars are superseded by the repeated-measures partial η^2 estimates in Sec. S16.

Table S2 preserves the complete per-dataset values from the original between-cell model. We do not use those significance tests because they assume independent cells. The valid repeated-measures effect sizes and tests are reported in Sec. S16.

Table S2: Original between-cell η^2 per model and dataset, retained for traceability only. Cov = η^2(coverage); Str = η^2(stride). See Sec. S16 for corrected inference.

Model	AUTSL		FineGym		SSv2		Diving-48		HMDB-51		DriveAct		EPIC		UCF-101	
	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str
VMamba	.88	.09	.77	.16	.88	.09	.94	.04	.90	.09	.83	.15	.95	.05	.88	.07
TSF	.92	.06	.74	.19	.87	.10	.97	.02	.92	.07	.84	.14	.91	.08	.85	.07
MC3	.66	.18	.50	.33	.54	.28	.81	.13	.81	.18	.75	.22	.89	.11	.79	.18
R3D	.60	.21	.47	.34	.60	.26	.77	.15	.68	.31	.58	.33	.75	.24	.70	.25
R2+1D	.63	.20	.50	.32	.62	.26	.76	.17	.72	.27	.49	.38	.80	.18	.76	.20
VMAE	.69	.18	.53	.28	.69	.22	.82	.12	.84	.16	.43	.43	.87	.11	.81	.14
ViViT	.61	.24	.53	.32	.51	.33	.69	.20	.71	.27	.60	.31	.80	.19	.68	.26
SF	.51	.29	.26	.50	.44	.36	.51	.33	.55	.39	.39	.46	.67	.30	.61	.36

S4 Action Sensitivity Taxonomy

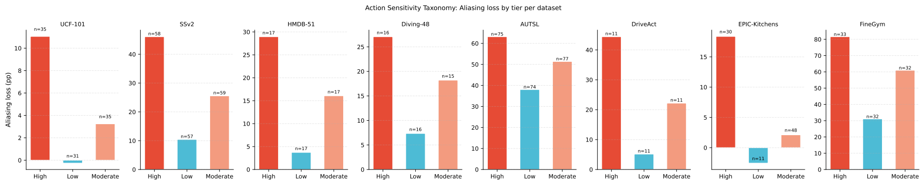


Fig. S11: Evidence loss (pp) per sensitivity tier and dataset. Each bar shows mean accuracy drop (stride=1→16) within the tier, annotated with the number of action classes. UCF-101 Low tier: −0.3pp; AUTSL remains high across tiers.

Table S3 lists the threshold values that define tier boundaries per dataset, derived from the 33rd and 67th percentiles of the per-class accuracy drop.

Table S3: Taxonomy tier thresholds (accuracy drop, pp): classes below the Low threshold alias negligibly; classes above the High threshold are severely affected. Thresholds are dataset-specific and computed across all valid architectures.

Dataset	Classes	Low threshold	High threshold	High count	Low count
AUTSL	226	66.9pp	73.9pp	75	74
Diving-48	47	25.0pp	47.7pp	16	16
SSv2	174	59.7pp	76.9pp	58	57
HMDB-51	51	8.4pp	28.5pp	17	17
EPIC-Kitchens	89	0.0pp	19.4pp	30	11
DriveAct	33	14.8pp	51.0pp	11	11
UCF-101	101	1.4pp	7.5pp	35	31

S5 Maximum-Probability Confidence-Cascade Curves — All Models

Figure S12 shows the descriptive in-sample accuracy–frame curves obtained by thresholding maximum class probability. These curves are not held-out estimates; held-out calibration results are reported in Sec. S17. Each curve sweeps $\tau \in [0.10, 0.95]$. Dashed horizontal lines indicate fixed-budget baselines (4f, 16f) and the oracle.

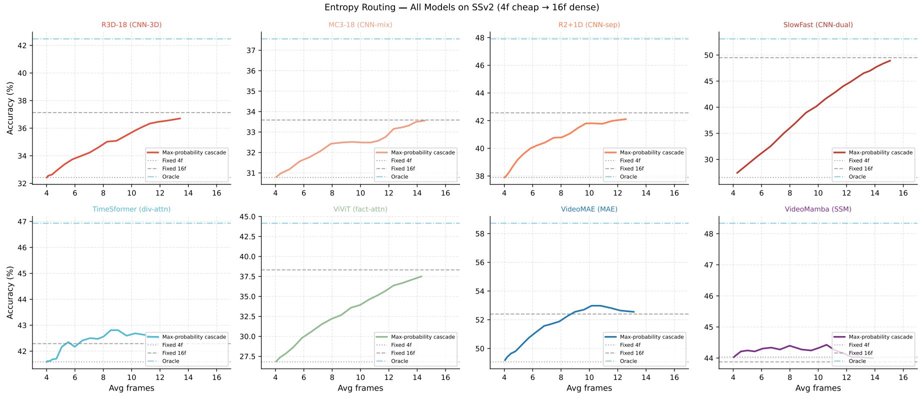


Fig. S12: Maximum-probability cascade: descriptive in-sample accuracy vs. average frames on SSv2 (4f cheap → 16f dense). Held-out threshold selection is analysed separately in Sec. S17.

Table S4 provides the full in-sample cascade frontier across all models and datasets at the optimal threshold (avg frames ≤ 8); it is retained as a descriptive diagnostic, not an unbiased performance estimate.

Table S4: In-sample maximum-probability cascade frontier: best Top-1 accuracy at avg ≤ 8 frames per model/dataset. Held-out results and uncertainty are in Sec. S17.

Model	AUTSL	FineGym	Diving	SSv2	HMDB	DriveAct	EPIC	UCF	Avg
R3D-18	73.3	68.5	27.6	34.6	80.2	66.1	36.7	81.6	58.6
MC3-18	63.4	62.9	31.8	32.4	78.5	69.4	36.5	85.6	57.6
R2+1D	74.2	72.9	36.3	40.8	73.3	59.6	34.8	89.0	60.1
SlowFast	59.5	57.8	46.6	35.0	76.3	67.4	37.4	87.9	58.5
TimeSformer	67.1	66.8	38.9	42.5	80.6	68.5	32.4	91.3	**61.0**
ViViT	67.2	66.1	48.9	32.2	80.1	64.3	31.3	94.3	60.6
VideoMAE	**79.5**	**76.4**	**52.6**	**51.9**	**84.7**	**69.0**	**37.8**	**95.9**	**68.5**
VideoMamba	65.0	73.2	37.4	44.4	69.8	57.6	28.4	88.5	58.0

S6 Clip Duration Analysis

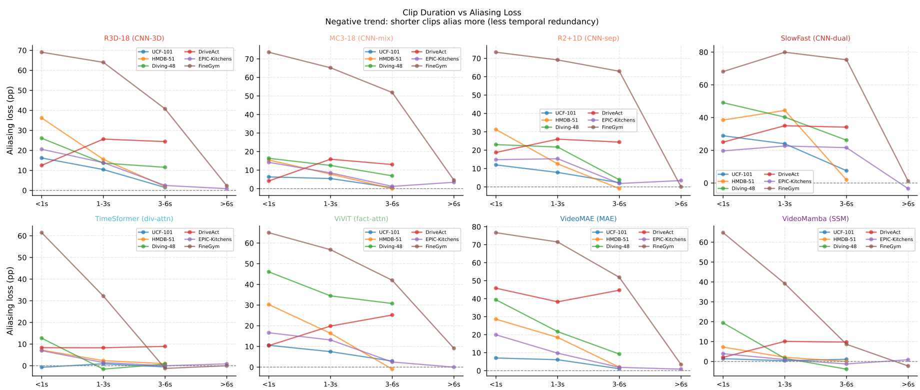


Fig. S13: Evidence loss (pp) vs. clip duration. Shorter clips lose more accuracy (Pearson $r = -0.3$ to -0.8 across models), consistent with lower temporal redundancy. Diving-48 is the exception.

S7 Exploratory Motion-Proxy Analysis

We estimated per-class temporal activity using optical-flow magnitude (Farneback algorithm [1]) computed on 5 representative videos per class, sampling 16 frames per video at 112×112px. This is an exploratory motion proxy, not a direct measurement of action frequency or a validation of classical Nyquist aliasing.

Table S5 reports Pearson correlation between per-class mean flow magnitude and evidence loss (accuracy drop stride=1→16). The range $r = -0.475$ to 0.327 shows that whole-frame motion is an inconsistent proxy: camera motion can dominate flow, localized hand motion can be diluted, and a brief discriminative event need not have high mean motion.

Table S5: Pearson r between whole-frame optical-flow magnitude and per-class evidence loss. The mixed signs show that flow magnitude is not a literal Nyquist proxy.

Dataset	Pearson r	p-value	Significant ($p<0.05$)
DriveAct	0.327	0.065	Marginal
AUTSL	0.285	<0.001	Yes
SSv2	0.184	<0.001	Yes
HMDB-51	0.180	0.206	No
Diving-48	0.140	0.349	No
UCF-101	0.031	0.762	No
EPIC-Kitchens	−0.002	0.985	No
FineGym	−0.475	<0.0001	Yes (inverted)

S8 Implementation Details

Hardware. Training experiments used NVIDIA A100 (40GB) and H200 NVL (141GB) GPUs (CUDA 13.2). Temporal-sweep evaluation (1,600 configurations) was parallelized across 8 nodes using a custom Slurm auto-submitter. Inference latency measurements used an NVIDIA RTX PRO 6000 Blackwell GPU (96GB) with bfloat16 autocast, batch size=1, 20 warmup iterations, 100 benchmark iterations, and CUDA event timing with explicit synchronization.

Splits, clip sampling, and evaluation determinism. We use the official train/validation/test split when a dataset provides one. When the public split is larger than needed for the controlled sweep, we form a deterministic balanced evaluation subset with approximately 20 validation clips per class; if a class has fewer available clips, all available validation clips are used. The same clip list is reused across all architectures, resolutions, coverages, and strides for a dataset. This makes pairwise stride/resolution differences comparable and avoids model-specific sampling noise. Reported main-paper numbers are validation-set measurements; no test-set tuning is used.

Seeds and uncertainty. Because the full grid requires thousands of model–dataset–resolution evaluations, each model–dataset fine-tuning run uses one fixed training seed and deterministic evaluation. Confidence intervals in S11 are therefore not random-seed intervals: they bootstrap over the 8-architecture pool. Per-class taxonomy and Levene analyses use class-level variation, and routing curves use validation-set threshold sweeps rather than an independently calibrated test threshold.

Frame and crop normalization. Training uses random resized crop, horizontal flip, and ImageNet normalization. Evaluation uses deterministic resize/center-crop at the requested spatial resolution. For every coverage/stride setting, frames are selected from the same leading candidate window. Each architecture receives its expected input length; if a candidate window is too short, we pad by repeating the last available frame, so stride changes alter sampling density rather than tensor shape.

Training hyperparameters. All models fine-tuned for 10 epochs from Kinetics-400 checkpoints. Transformers and VideoMamba: AdamW, lr=2×10^{-5}, weight decay=0.05, batch size=32 per GPU. CNNs: AdamW, lr=10^{-4}, batch size=128 per GPU. SlowFast: AdamW, lr=10^{-4}, batch size=32 per GPU. All models use cosine learning rate decay with linear warmup (1 epoch). Spatial augmentation: random resized crop, horizontal flip, normalization (ImageNet statistics).

Positional embedding interpolation for Transformers and SSMs. For target-resolution fine-tuning at non-native resolutions, Transformer and SSM positional embeddings must be adapted. *The wrong approach*: using the HuggingFace mismatched-size flag discards positional embeddings entirely, causing a large accuracy cliff (typically 10–20pp worse than our approach). *Our approach*: load the native-resolution checkpoint, then bicubically interpolate spatial positional embeddings from the native grid (14×14=196 tokens at 224px) to the target grid ($\lfloor r/16 \rfloor^2$ tokens at resolution r) before copying weights to a model instantiated at the target resolution. This preserves the spatial topology learned during pre-training and ensures fine-tuning starts from a meaningful positional encoding rather than a random one. The same bicubic interpolation applies to TimeSformer, VideoMAE, ViViT, and VideoMamba.

SlowFast spatial pooling fix. The original SlowFast-R50 head uses a fixed 7×7 average pooling kernel, assuming 224px input. For non-native resolutions, we replace both head pooling layers with `AdaptiveAvgPool3d(1)`, enabling forward passes at arbitrary input resolutions without parameter changes.

FineGym dataset details. FineGym [2] contains 97 fine-grained gymnastics action classes (floor exercise, uneven bars, balance beam, vault; parallel bars, horizontal bar, pommel horse, rings for men). We use the FineGym288 subset (1,960 clips), with deterministic balanced clip lists for the controlled sweeps. Preprocessing: clips are trimmed to the annotated action boundary (mean duration 3.2s at 25fps) and padded with the last frame if shorter than 16 frames.

S9 Multi-Dataset Spatial Resolution Results

Tables S6–S13 extend the spatial robustness analysis from the main paper (Section 4.3) to all 8 datasets. Tables S6–S12 show spatial-resolution results at 96–224px. Table S13 shows FineGym target-resolution fine-tuning results at 48–224px (one fine-tuned checkpoint per resolution). **Bold** = native training resolution. $\Delta_{\max}$ = largest accuracy drop from native across all tested resolutions.

For the no-fine-tuning tables, “resolution” denotes the square pre-resize applied to decoded source frames before they are passed through the unchanged native checkpoint and its stored preprocessing. These are therefore input-scale/resampling stress tests, not evaluations of a Transformer instantiated on a new positional-token grid. Positional embedding interpolation is used only in the target-resolution fine-tuning branch.

Table S6: Input-scale sensitivity on **SSv2** under source-frame pre-resizing and unchanged native checkpoint preprocessing (repeated from main paper; no fine-tuning).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	35.9	**37.1**	30.3	17.2	−19.9pp
MC3-18	112px	32.4	**33.6**	27.8	14.8	−18.8pp
R2+1D	112px	40.1	**42.6**	36.2	20.8	−21.8pp
SlowFast	224px	27.7	32.4	45.7	**49.5**	−21.8pp
TimeSformer	224px	39.3	39.4	41.4	**42.3**	−3.0pp
ViViT	224px	36.4	37.1	38.1	**38.3**	−1.9pp
VideoMAE	224px	48.9	49.2	51.5	**52.3**	−3.4pp
VideoMamba	224px	37.5	39.3	42.5	**43.9**	−6.4pp

Table S7: Spatial robustness on **UCF-101** (no retraining).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	79.9	**81.2**	75.3	55.1	−26.1pp
MC3-18	112px	82.9	**85.4**	81.0	63.9	−21.5pp
R2+1D	112px	88.0	**88.6**	85.1	64.9	−23.7pp
SlowFast	224px	69.7	75.8	87.5	**91.4**	−21.7pp
TimeSformer	224px	89.1	89.7	90.7	**91.5**	−2.4pp
ViViT	224px	92.0	93.5	94.7	**95.1**	−3.1pp
VideoMAE	224px	95.3	95.7	96.1	**96.4**	−1.1pp
VideoMamba	224px	76.7	78.7	82.6	**83.5**	−6.8pp

Table S8: Spatial robustness on **HMDB-51** (no retraining).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	78.8	**80.3**	75.9	59.8	−20.5pp
MC3-18	112px	76.7	**78.6**	74.2	51.0	−27.6pp
R2+1D	112px	74.4	**73.1**	68.4	50.5	−22.6pp
SlowFast	224px	59.5	67.2	80.5	**81.6**	−22.1pp
TimeSformer	224px	75.9	77.2	77.5	**78.9**	−3.0pp
ViViT	224px	78.0	79.0	79.9	**79.5**	−1.5pp
VideoMAE	224px	83.7	83.1	84.0	**83.8**	−0.7pp
VideoMamba	224px	59.9	63.8	67.2	**69.8**	−9.9pp

Table S9: Spatial robustness on **Diving-48** (no retraining).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	26.5	**28.8**	22.1	12.8	−16.0pp
MC3-18	112px	30.8	**31.6**	27.3	13.7	−17.9pp
R2+1D	112px	33.8	**35.3**	30.1	18.7	−16.6pp
SlowFast	224px	18.4	25.0	43.3	**50.5**	−32.1pp
TimeSformer	224px	37.4	39.3	38.6	**38.1**	−0.7pp
ViViT	224px	48.0	49.9	52.8	**53.0**	−5.0pp
VideoMAE	224px	48.7	48.3	50.9	**48.6**	−0.3pp
VideoMamba	224px	29.2	29.1	36.1	**36.6**	−7.5pp

Table S10: Spatial resolution results on **AUTSL** (sign language). CNN degradation is severe under off-native evaluation; SlowFast drops from 79.2% at 224px to 22.5% at 96px (−56.7pp).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	72.2	**75.0**	63.0	27.9	−47.1pp
MC3-18	112px	59.1	**63.7**	52.7	12.2	−51.5pp
R2+1D	112px	73.7	**75.9**	68.9	38.4	−37.5pp
SlowFast	224px	22.5	32.6	70.0	**79.2**	−56.7pp
TimeSformer	224px	65.1	66.0	68.3	**68.5**	−3.4pp
ViViT	224px	49.2	51.8	53.3	**53.6**	−4.4pp
VideoMAE	224px	77.3	78.2	79.5	**79.6**	−2.3pp
VideoMamba	224px	30.4	34.2	53.3	**65.8**	−35.4pp

Table S11: Spatial robustness on **DriveAct** (in-cabin activities; no retraining).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	65.0	**68.3**	52.7	19.0	−49.3pp
MC3-18	112px	65.2	**69.0**	38.4	6.2	−62.8pp
R2+1D	112px	67.4	**62.5**	40.0	12.9	−49.6pp
SlowFast	224px	31.7	40.2	65.8	**70.1**	−38.4pp
TimeSformer	224px	68.3	69.2	69.6	**70.3**	−2.0pp
ViViT	224px	59.4	58.7	62.1	**63.4**	−4.7pp
VideoMAE	224px	71.4	69.9	71.7	**73.0**	−3.1pp
VideoMamba	224px	35.0	38.2	46.9	**57.8**	−22.8pp

Table S12: Spatial robustness on **EPIC-Kitchens** (egocentric cooking; no retraining).

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	33.3	**37.2**	26.7	8.6	−28.6pp
MC3-18	112px	36.1	**36.2**	27.2	9.4	−26.8pp
R2+1D	112px	32.8	**35.5**	27.6	11.8	−23.7pp
SlowFast	224px	11.3	17.7	31.3	**37.5**	−26.2pp
TimeSformer	224px	20.5	19.2	25.6	**31.7**	−12.5pp
ViViT	224px	30.2	34.1	36.1	**37.9**	−7.7pp
VideoMAE	224px	30.9	32.6	36.5	**38.6**	−7.7pp
VideoMamba	224px	11.1	12.6	20.8	**28.3**	−17.2pp

Table S13: FineGym spatial resolution results after target-resolution fine-tuning (one checkpoint per resolution, 10 fine-tuning epochs from native checkpoint). Bold = native training resolution (112px for CNNs, 224px for Transformers/SSM). Unlike other datasets in S9 (cross-resolution without retraining), FineGym shows CNN performance that *improves beyond native* after fine-tuning at higher resolutions (R3D-18: 71.5%@112px→76.1%@224px), consistent with higher spatial resolution aiding gymnastics pose discrimination.

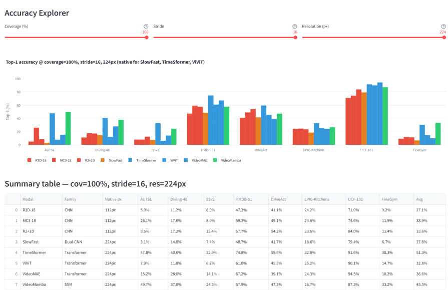
Model	Native	48px	96px	112px	160px	224px
R3D-18	112px	56.4	72.6	**71.5**	74.9	76.1
MC3-18	112px	46.3	62.3	**64.3**	73.8	65.7
R2+1D	112px	61.6	73.1	**76.8**	79.4	76.2
SlowFast	224px	36.3	67.2	69.6	76.3	**78.2**
TimeSformer	224px	24.3	48.9	50.8	62.2	**66.4**
ViViT	224px	21.3	39.1	40.8	49.6	**69.9**
VideoMAE	224px	25.4	51.1	57.5	64.8	**78.0**
VideoMamba	224px	29.1	51.6	58.2	66.0	**73.2**

S10 Interactive Dashboard and Code Availability

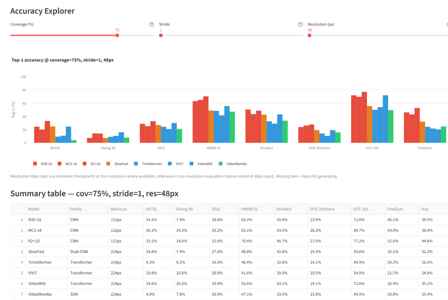
Interactive visualization dashboard. An anonymized interactive dashboard is included with the supplementary material for review. The public dashboard URL will be released after the review period. The dashboard provides:

1. **Temporal explorer**: coverage×stride heatmaps per model and dataset, with configurable scaling and stride annotations.
2. **Spatial browser**: accuracy curves across resolutions for all 8 models and datasets; target resolution tuning by family.
3. **Recommender**: Given a target dataset, latency budget, frame rate, and deployment context, the review package applies deterministic rules from the paper tables to recommend a model–resolution–stride operating point. Any optional natural-language summarizer is disabled by default and is not used for any scientific result.
4. **TDS comparison**: rank any subset of the 8 datasets by temporal demand, with model-specific breakdowns and Spearman correlation display.
5. **Routing simulator**: maximum-probability threshold sweep for any model on any dataset, with visual Pareto-frontier display.

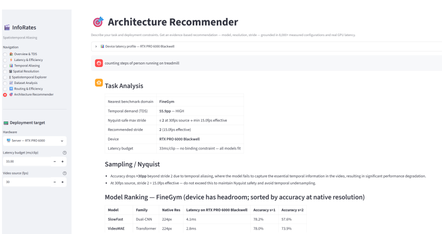
Code availability. Evaluation scripts, model fine-tuning code, result tables, and the Streamlit dashboard source are provided as an anonymized supplementary package for review. This includes the target-resolution pipeline with bicubic positional embedding interpolation. The public repository will be released after review.



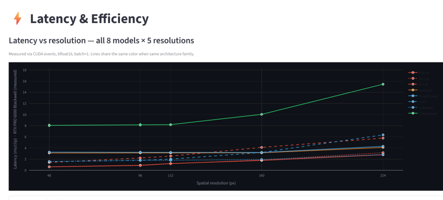
(a) Accuracy Explorer: native resolution (stride=16, cov=100%, 224px). Summary table shows top-1 across all 8 datasets.



(b) Accuracy Explorer: low resolution (stride=1, cov=75%, 48px). Retrained CNN checkpoints are used at 48px.



(c) Architecture Recommender: deterministic FineGym example grounded only in the packaged result tables.



(d) Latency & Efficiency: measured forward-pass latency (ms/clip, bfloat16, RTX PRO 6000) vs. spatial resolution for all 8 architectures. VideoMamba (green) is 4–8× slower than VideoMAE across all resolutions.

Fig. S14: Interactive dashboard screenshots from the anonymized supplementary package. The Accuracy Explorer (a,b) lets users sweep coverage, stride, and resolution interactively and view per-dataset top-1 accuracy tables. The Architecture Recommender (c) uses deterministic rules grounded in packaged measurements; no external API is required. The Latency & Efficiency tab (d) displays GPU latency curves across all model–resolution pairs.

S11 Leave-One-Architecture-Out TDS Stability

To assess whether the TDS ranking depends on the specific 8-architecture pool evaluated, we recompute TDS for all 8 datasets after removing each architecture in turn (7-architecture pools) and compare the resulting ranking to the full 8-architecture ranking via Spearman ρ.

Table S14: Leave-one-architecture-out TDS ranking stability. ρ is the Spearman correlation between the 8-dataset ranking computed with the named architecture excluded (7-architecture pool) and the full 8-architecture ranking.

Excluded architecture	Spearman ρ	p
R3D-18	1.000	$< 10^{-4}$
MC3-18	1.000	$< 10^{-4}$
R2+1D	1.000	$< 10^{-4}$
SlowFast	1.000	$< 10^{-4}$
TimeSformer	1.000	$< 10^{-4}$
ViViT	1.000	$< 10^{-4}$
VideoMAE	1.000	$< 10^{-4}$
VideoMamba	0.976	3.3×10^{-5}

For 7 of 8 leave-one-out pools, the ranking is *exactly* identical to the full-pool ranking ($\rho = 1.00$). One leave-one-out pool swaps only the #1/#2 positions ($\rho = 0.976$); ranks 3–8 are unaffected by every exclusion. This is sensitivity of the architecture-averaged estimator, not a protocol difference: every model–dataset pair is processed identically. We therefore report FineGym and AUTSL as co-equal highest-TDS domains rather than asserting a strict order.

S11.1 Architecture-Family Ablation

Leave-one-out removes a single architecture at a time; a stronger test removes an entire family. We recompute TDS under three restricted pools: **CNN-only** (n=4: R3D-18, MC3-18, R2+1D, SlowFast), **Transformer-only** (n=3: TimeSformer, ViViT, VideoMAE), and **Transformer+SSM** (n=4: adding VideoMamba). Table S15 reports TDS under each pool and Spearman ρ against the full ranking.

The ranking is essentially invariant to architecture-family composition: removing all four CNNs, keeping only the three Transformers, or keeping Transformers+SSM all leave ranks 3–8 in identical order, and only the CNN-only pool reproduces the same FineGym/AUTSL swap already characterized in the leave-one-out test above ($\rho = 0.976$ in both cases, for the same reason: CNNs are comparatively more fragile on AUTSL's fine handshape transitions than on FineGym's phase boundaries, which narrows the gap). This directly addresses whether TDS reflects a property of the dataset pool versus an artifact of including any one architecture family: under every family-restricted pool we tested,

Table S15: TDS per dataset under architecture-family ablation pools, compared to the full 8-architecture ranking (sorted by full-pool TDS).

Dataset	Full (n=8)	CNN-only (n=4)	Transf.-only (n=3)	Transf.+SSM (n=4)
FineGym	55.92	62.05	53.06	49.80
AUTSL	53.02	67.20	46.45	38.83
SSv2	27.29	32.16	25.95	22.43
DriveAct	21.48	24.05	23.21	18.92
Diving-48	19.16	21.74	20.59	16.59
HMDB-51	16.47	21.47	14.54	11.47
EPIC-Kitchens	9.77	12.89	8.27	6.64
UCF-101	4.86	7.36	2.90	2.35
Spearman ρ vs. full	—	0.976	1.000	1.000

the ranking is the same dataset ordering reported in the main paper, with the sole exception of the already-disclosed #1/#2 ambiguity.

S11.2 Bootstrap Confidence Intervals

To quantify ranking uncertainty directly rather than only testing discrete leave-one-out/family pools, we bootstrap over architectures: we resample 8 architectures with replacement (B=10,000 resamples) from the original pool, recompute TDS for each dataset on each resample, and report the resulting 95% percentile confidence interval in Table S16.

Table S16: Bootstrap 95% CI for TDS per dataset (resampling 8 architectures with replacement, B=10,000).

Dataset	TDS (bootstrap mean)	95% CI
FineGym	55.99	[47.24, 63.86]
AUTSL	53.13	[36.02, 67.03]
SSv2	27.35	[20.40, 33.92]
DriveAct	21.57	[13.92, 29.35]
Diving-48	19.23	[10.67, 28.37]
HMDB-51	16.54	[9.53, 24.21]
EPIC-Kitchens	9.80	[6.25, 13.22]
UCF-101	4.88	[2.18, 8.36]

All eight 95% CIs are disjoint from zero, indicating non-zero temporal demand at the architecture-pool level. AUTSL's CI is the widest (width 31.0pp) because its per-architecture drops are the most heterogeneous (TimeSformer and VideoMamba near-flat, CNNs collapse sharply), while EPIC-Kitchens and UCF-101 have the narrowest CIs, consistent with their low and relatively architecture-independent TDS.

We additionally bootstrap the FineGym−AUTSL gap directly (same resampling procedure, computing TDS(FineGym)−TDS(AUTSL) on each resample):

mean = +2.92pp, 95% CI = [−4.58, +11.73], with FineGym exceeding AUTSL in 73.9% of resamples. The confidence interval includes zero, which is the appropriate statistical test of the co-equal-ranking claim: the data are consistent with FineGym narrowly leading AUTSL in expectation, but do not support rejecting the null hypothesis of equal TDS at the 95% level. This is the quantitative basis, rather than only the leave-one-out swap, for reporting FineGym and AUTSL as co-equal highest-TDS domains in the main paper.

S12 Spectral and Evidence-Localization Checks

The main paper no longer interprets TDS as literal signal aliasing: it measures accuracy degradation under the implemented stride and coverage protocol. Here we report two diagnostics. The whole-frame optical-flow spectral proxy fails in the opposite direction from a naive signal-level Nyquist prediction; the second asks whether classifier evidence is temporally localized. Both are exploratory label-space diagnostics, not classical spectral validation.

Methodology. For each of 8 datasets and 5 resolutions (48, 96, 112, 160, 224 px), we sample up to 12 classes × 2 videos. We decode up to 64 *consecutive* frames in native order, resize to the target resolution, and compute inter-frame optical-flow magnitude (Farneback) as a temporal signal sampled once per original frame. We estimate each signal's spectral cutoff frequency f_c (90% cumulative periodogram power, cycles/frame) and average over sampled videos to obtain one f_c per (dataset, resolution) pair. We pair each f_c with empirical stride-sensitivity at the same resolution: mean accuracy drop from stride=1 to stride=16 at coverage=100%, averaged over 8 architectures and read from the existing sweep. This yields $8 \times 5 = 40$ (dataset, resolution) points.

Table S17: Mean flow-derived spectral cutoff frequency f_c, videos per resolution, and mean empirical stride-sensitivity, averaged over the 5 tested resolutions per dataset and sorted by stride-sensitivity.

Dataset	Mean f_c	n/res.	Mean stride-sens.
FineGym	0.188	24	46.30
AUTSL	0.121	24	45.61
SSv2	0.339	24	25.70
DriveAct	0.192	21	18.65
HMDB-51	0.383	24	16.57
Diving-48	0.182	24	15.21
EPIC-Kitchens	0.248	23	9.70
UCF-101	0.387	24	5.45

Result. Pooling across all 40 (dataset, resolution) points, f_c is significantly *negatively* correlated with stride-sensitivity (Spearman $\rho = -0.635$, permutation

$p = 1.0 \times 10^{-5}$, $n = 40$)—the opposite sign from the naive prediction that higher whole-frame temporal-frequency content should require denser sampling. FineGym integrates cleanly into this pattern: it has the highest mean stride-sensitivity (46.30pp) but a moderate whole-frame cutoff frequency ($f_c = 0.188$ cycles/frame), close to DriveAct and Diving-48 rather than UCF-101/HMDB-51.

Interpretation. The FineGym inversion is part of a broader descriptive pattern across the sampled dataset–resolution points. UCF-101 and HMDB-51—appearance- and sports-driven domains with large whole-body and camera motion—have the highest flow-derived cutoff frequencies but among the lowest stride-sensitivity, because the action remains identifiable from almost any sampled frame. AUTSL and FineGym show the opposite behavior: their discriminative evidence is localized in handshape changes or brief gymnastics phase events, which are diluted by a whole-frame flow aggregate. In this sample, whole-frame optical-flow frequency is therefore an inverted proxy for classifier evidence demand, not a validation of a spectral mechanism.

Label-Space Evidence-Localization Checks

The inversion above arises because whole-frame optical flow measures *pixel-level motion magnitude*, not the temporal localisation of *discriminative evidence*. We conduct three complementary tests on FineGym ($n = 97$ classes) using the fine-tuned TimeSformer checkpoint, which probe whether the model's evidence is temporally concentrated.

Test A: Temporal burstiness of optical flow (CPU). Per-class, we compute the temporal coefficient of variation of inter-frame flow magnitudes (CV $= \sigma_{\text{time}}/\mu_{\text{time}}$). After controlling for mean flow, temporal standard deviation positively predicts evidence loss: $\partial r = +0.235$, $p = 0.021$. The strong correlation between standard deviation and mean flow ($r = +0.909$) limits mechanistic interpretation.

Test B: TimeSformer temporal attention entropy (GPU). We register forward hooks on the temporal self-attention Dropout layer in all 12 TimeSformer blocks, capturing the $[196 \times 12 \times 8 \times 8]$ temporal attention weight tensor per clip. Mean attention entropy positively correlates with evidence loss: $r = +0.317$, $p = 0.0015$. This is an association, not a mechanism test.

Test C: Sliding-window confidence concentration (GPU, most direct). For each clip we decode 48 consecutive frames, slide a window of width $T = 8$ with stride 4, and record correct-class probability. The concentration $1 - H/\log n_{\text{windows}}$ measures how localized the model's evidence is. Concentration positively predicts evidence loss: $r = +0.263$, $p = 0.009$; controlling for mean optical-flow magnitude gives $\partial r = +0.385$, $p < 0.001$ ($n = 97$ classes).

Table S18: Label-space evidence-localization checks on FineGym ($n = 97$ classes). All three tests use a fine-tuned TimeSformer trained on FineGym. Partial r controls for whole-frame optical-flow magnitude.

Test	Feature	r	p	partial r	p
A: Optical flow burstiness	$\sigma_t(\text{flow})/\mu_t(\text{flow})$	-0.421	<0.001	$+0.235$	0.021
B: Temporal attention entropy	Mean attention H (nats)	$+0.317$	0.0015	—	—
C: Sliding-window confidence	Confidence concentration	$+0.263$	0.009	$+\mathbf{0.385}$	$<\mathbf{0.001}$

Summary. Table S18 collects the three results.

Test C provides the clearest classifier-level diagnostic: temporal concentration of correct-class confidence positively predicts per-class evidence loss. Classes whose evidence is confined to a narrow window are more likely to lose that window under the protocol. This does not establish literal spectral aliasing.

Generalisation of Test C to All 8 Datasets

We generalise Test C to all 8 datasets and an architecture ensemble. We correlate per-class confidence concentration with per-class evidence loss (mean accuracy drop stride=1→16) via Spearman ρ.

Window-geometry constraint. The metric requires the decoded sequence to be substantially longer than the window T, so that windows are well separated and concentration reflects genuine temporal localisation. With 48 decoded frames, models with $T \leq 16$ satisfy this (neighbouring-window overlap $\leq 81\%$), but the two models with $T = 32$ (ViViT, SlowFast) collapse to stride 1 and 97% overlap: their 17 windows differ by a single frame, the confidence profile is near-flat, and the concentration measure floors out (mean concentration 0.10 vs. 0.22–0.36 for $T \leq 16$ models). On short-clip datasets such as FineGym (median ≈37 source frames) a 32-frame window cannot be slid at all. We therefore restrict the ensemble to the 6 architectures with $T \leq 16$ (TimeSformer, VideoMAE, R3D-18, MC3-18, R2+1D, VideoMamba); the $T = 32$ models are a measurement artifact of the protocol rather than a property of the data.

Result. Table S19 reports positive correlations in all 8 datasets (pooled Spearman $\rho = +0.257$), significant in 4. We treat this as converging evidence for an association between temporal concentration and evidence loss, not as a causal or sampling-theoretic proof.

Generalisation of Test B (TimeSformer temporal attention entropy) to the remaining datasets requires per-dataset attention extraction and is left to future work; the sliding-window concentration test (Test C), being architecture-agnostic, is the one we generalise here.

Table S19: Sliding-window confidence-concentration test across all 8 datasets. Spearman ρ relates per-class confidence concentration to per-class evidence loss. The ensemble uses models with window $T \leq 16$; **bold** denotes $p < 0.05$.

Dataset	Classes	Models	Spearman ρ	p
HMDB-51	51	6	**+0.606**	<0.001
UCF-101	101	6	**+0.442**	<0.001
DriveAct	33	6	**+0.430**	0.013
SSv2	174	4	**+0.269**	<0.001
FineGym	97	6	+0.138	0.177
EPIC-Kitchens	89	6	+0.093	0.386
AUTSL	226	6	+0.074	0.270
Diving-48	47	6	+0.001	0.994
Pooled mean			**+0.257**	—

S13 Additional Robustness Checks

Observational architecture interpretation. The architecture comparison in the main paper is intentionally phrased as an empirical association rather than a controlled causal ablation. Attention layout, state-space recurrence, tokenization, pretraining objective, parameter count, input length, native resolution, and optimization all covary across the evaluated architectures. A definitive mechanism test would require holding the backbone fixed while swapping only temporal aggregation modules. Moreover, the original sweep is confounded by native frame budget. Our revised architecture result therefore comes only from the matched-evidence comparison in Sec. S15: an accuracy-per-distinct-frame ordering remains, but neither causality nor the original 3–5× magnitude is supported.

Dense-accuracy headroom does not explain TDS. Because several datasets use balanced subsets or fixed 20-clips-per-class protocols, external leaderboard gaps are not directly comparable. We therefore compute internal dense-accuracy headroom. If under-optimization drove evidence loss, headroom should correlate positively with stride=1→16 drop. It does not: across 64 model–dataset pairs, Spearman $\rho = -0.22$ ($p = 0.079$); at dataset level, mean headroom vs. TDS gives $\rho = 0.36$ ($p = 0.385$).

Coverage–stride interaction. Figures in S1 visualize the complete coverage×stride grids. To test additivity, we average accuracy over architectures for one high-TDS and one low-TDS dataset, then decompose each 5×5 grid into row/column effects plus an interaction residual. FineGym has a meaningful interaction component: 15.0% of grid variance, with RMSE 8.3pp. Low coverage is poor regardless of stride, while high coverage exposes the stride cliff. UCF-101 is closer to additive (4.0%; RMSE 1.7pp).

Confidence-cascade calibration. The cascade thresholds maximum class probability, not predictive entropy. We select τ on a stratified calibration half and score it on the untouched half over 1000 resamples. On SSv2/TimeSformer the

Table S20: Dense-accuracy headroom check. Headroom is the mean gap from each model's stride=1 accuracy to the best stride=1 accuracy achieved by any model on the same dataset. It is not significantly correlated with TDS, suggesting that temporal sensitivity is not an artifact of under-optimized dense accuracy.

Dataset	Mean stride=1	Best stride=1	Mean headroom	TDS
FineGym	73.8	79.9	6.1	55.9
AUTSL	72.9	82.3	9.3	53.0
SSv2	42.5	52.4	9.9	27.6
DriveAct	67.4	73.9	6.5	21.9
Diving-48	40.3	53.0	12.7	19.2
HMDB-51	78.2	84.0	5.8	16.6
EPIC-Kitchens	34.9	39.4	4.5	9.7
UCF-101	89.0	95.5	6.5	4.9

Table S21: Mean accuracy (%) over architectures for a high-TDS dataset (FineGym) and a low-TDS dataset (UCF-101). Rows are coverage; columns are stride.

FineGym						UCF-101					
Coverage	s1	s2	s4	s8	s16	Coverage	s1	s2	s4	s8	s16
10%	8.0	7.2	6.4	5.8	5.8	10%	74.5	73.2	69.5	64.2	57.7
25%	16.9	15.6	12.4	8.3	6.4	25%	81.7	81.6	80.1	75.0	67.8
50%	42.9	40.7	28.7	17.1	9.3	50%	86.6	86.6	85.7	82.9	76.6
75%	63.4	59.6	42.3	25.1	13.1	75%	88.4	88.7	88.5	86.8	81.4
100%	73.8	68.0	49.8	29.8	15.7	100%	89.0	89.0	89.0	88.1	84.1

held-out change relative to dense inference is -0.03pp (95% CI $[-1.07, +0.83]$) at 6.9 average frames. Across 56 model–dataset pairs, however, the mean change is -3.5pp. The result supports confidence as a useful signal for some pairs, not a universally accuracy-neutral routing policy.

S14 What the Stride Axis Varies

[New in this revision, in response to the reviews.]

Reviewers asked whether our temporal sweep measures aliasing or evidence loss. This section documents the sampler's behaviour and the measurements that settle the question.

Sampler. For a clip of T frames at coverage $\mathcal{C}$ and stride s, with model frame budget B, the candidate pool is $\{0, s, 2s, \dots\}$ within the first $\lfloor T\mathcal{C}/100 \rfloor$ frames. If the pool holds at least B entries it is re-uniformised to exactly B by linear interpolation over its index range; otherwise the shortfall is filled by repeating the last candidate. Table S22 makes both regimes concrete.

Sampling interval in seconds. Expressing the axis physically (Table S23) shows the same thing without reference to the implementation: over most of its range the stride axis does not change the rate at which the clip is sampled. Frame

Table S22: Frame positions actually supplied to a B=16 model at $\mathcal{C}$=100%. For a long clip the selected positions are nearly independent of stride; for a short clip the input degenerates into a few distinct frames followed by a static tail.

Clip	s	Distinct	Positions (last six)
T=400	1	16	266, 293, 319, 346, 372, 399
T=400	4	16	264, 292, 316, 344, 368, 396
T=400	16	16	256, 288, 304, 336, 352, 384
T=60	1	16	39, 43, 47, 51, 55, 59
T=60	8	8	56, 56, 56, 56, 56, 56
T=60	16	4	48, 48, 48, 48, 48, 48

rates here are measured from the files, not assumed; three datasets differ from the values used in our earlier duration analysis (HMDB-51 is 30 not 25 fps, AUTSL 30 not 25, EPIC-Kitchens 50 not 60).

Table S23: Median seconds between sampled frames at $\mathcal{C}$=100%. At a fixed nominal stride the realised interval differs by up to 3.9× across datasets, so equal-stride comparisons are not equal-rate comparisons.

Dataset	fps	s=1	s=2	s=4	s=8	s=16
UCF-101	25	0.395	0.395	0.395	0.395	0.617
Diving-48	25	0.265	0.265	0.265	0.314	0.600
SSv2	12	0.255	0.255	0.325	0.620	1.167
DriveAct	15	0.183	0.183	0.256	0.500	1.000
HMDB-51	30	0.167	0.167	0.167	0.260	0.504
AUTSL	30	0.125	0.125	0.131	0.252	0.475
EPIC-Kitchens	50	0.120	0.120	0.120	0.158	0.301

The effect is confined to the shortfall regime. Restricting to clips whose candidate pool never falls below B at s=16, the stride 1 → 16 accuracy difference is −0.51pp on average, against +10.74pp over all clips. This holds in **21 of 21** model–dataset pairs with at least 20 such clips. On AUTSL, SSv2 and DriveAct no clip escapes the shortfall regime at s=16, so the entire reported effect on those datasets is measured inside it. The unpadded subset is necessarily the longer clips, so this comparison is length-confounded on its own; it should be read together with the index and interval tables above, which do not depend on any subsetting.

Relation to TDS. Across datasets, the fraction of clips in the shortfall regime correlates with TDS at Spearman 0.964. The relationship is gating rather than explanatory: among AUTSL, SSv2 and DriveAct, all of which are fully saturated, TDS still spans 21.9–53.0pp. Clip length relative to frame budget determines whether temporal demand is expressed at all; dataset content determines how large it is.

S15 Matched-Evidence Architecture Comparison

[New in this revision, in response to the reviews.]

Because frame budget B governs when each architecture enters the shortfall regime, and B is rank-correlated with the reported drop at Spearman 0.849 (p=0.008), the uncontrolled comparison cannot isolate temporal aggregation design. We therefore supply every architecture the same k distinct frames at the same positions, uniformly spanning the clip, resampled to each model's own input length by uniform interpolation rather than end-repetition. Only $k \leq 8$ is strictly matched, since the smallest budget in the pool is 8.

Table S24 reports the complete matched results over the seven datasets with outputs for all eight architectures. Rankings are moderately stable between adjacent budgets (Spearman 0.69–0.83). At k=2, accuracy agrees with the ordering under the original protocol (Spearman 0.857, p=0.007); the correlations at k=1, 4, 8 are 0.643, 0.619, and 0.476 and are not significant. Ranked instead by the absolute accuracy drop between k=8 and k=2, the ordering does not reproduce the original one (Spearman 0.14, p=0.76).

Table S24: Matched-evidence Top-1 accuracy (%), averaged over the seven datasets with complete outputs for all eight architectures. Every model receives the same k distinct frame positions. $\Delta_{8\to2}$ is the absolute accuracy drop from $k = 8$ to $k = 2$.

Model	$k = 1$	$k = 2$	$k = 4$	$k = 8$	$\Delta_{8\to2}$
TimeSformer	21.1	33.8	52.1	58.6	24.8
VideoMamba	21.9	28.4	41.0	44.5	16.1
MC3-18	24.7	29.5	39.8	51.0	21.6
R3D-18	18.2	24.8	36.6	50.1	25.4
R2+1D	18.0	25.3	37.5	50.5	25.3
ViViT	20.4	27.6	41.3	53.6	25.9
VideoMAE	17.5	23.2	38.2	48.7	25.5
SlowFast	18.4	24.1	32.6	38.5	14.4

The budget confound is removed by construction. Consistent with frame budget being a major confound, the rank correlation between input length and measured degradation falls from 0.849 (p=0.008) under the published protocol to 0.231 (p=0.58) under matched evidence. The spread in absolute drop narrows from 4.1× under the original protocol to 1.8× under matched evidence. We therefore withdraw the original magnitude and mechanism claims; the matched experiment supports only a model-level evidence-efficiency comparison.

All eight architectures are included under the same protocol.

S16 Repeated-Measures Analysis of the Coverage×Stride Grid

[New in this revision, in response to the reviews.]

All 25 cells of the grid score the same clips, so treating them as independent observations, as our original two-way ANOVA did, understates the error term. We re-estimate with clip as subject and coverage and stride as fully crossed within-subject factors, testing each effect against its own subject×factor interaction and reporting partial η^2. Only clips present in all 25 cells are used, so the design stays balanced.

Over 56 model–dataset pairs, partial η^2 is 0.292 ± 0.152 for coverage, 0.232 ± 0.145 for stride and 0.070 ± 0.081 for their interaction. Coverage therefore remains dominant but by 1.26× rather than the 2.2× implied by the between-cell analysis. The interaction, which the original model omitted entirely, is largest on the high-demand datasets, reaching 0.271 on SlowFast/AUTSL, 0.262 on R3D-18/AUTSL and 0.232 on SlowFast/FineGym. The per-model ordering of the stride effect is unchanged under the corrected model (TimeSformer 0.124, MC3-18 0.197, R3D-18 0.226, R(2+1)D 0.238, ViViT 0.247, VideoMAE 0.259, SlowFast 0.331), so the qualitative grouping in the main paper does not depend on the error term.

S17 Held-Out Evaluation of the Confidence Cascade

[New in this revision, in response to the reviews.]

The cascade thresholds the maximum class probability, not the predictive entropy; our earlier name for it was inaccurate. More substantively, the threshold was previously selected and scored on the same split. We repeat the evaluation with τ chosen on a stratified calibration half and scored on the untouched half, over 1000 resamples.

On SSv2 with TimeSformer, the advantage over fixed dense inference falls from +0.21pp in-sample to −0.03pp held-out, with a 95% interval of $[-1.07, +0.83]$: the cascade matches dense accuracy at 6.9 average frames rather than exceeding it. That cell is also the most favourable one. Averaged over all 56 model–dataset pairs the held-out change relative to fixed dense inference is −3.5pp, and 25 of 56 pairs have intervals excluding zero, with losses reaching −14.7pp (ViViT/FineGym) and −10.5pp (VideoMAE/FineGym). The supportable claim is that confidence carries usable routing signal on some model–dataset pairs, not that temporal budgets can be halved without cost.

S18 Convergence of Low-Resolution Fine-Tuning

[New in this revision, in response to the reviews.]

To check whether the 10-epoch budget limits the reported off-native accuracies, we recovered the validation curves available at the five reported target resolutions (n=503 logs) and asked whether each run was still improving when the budget ended.

The concern is partly borne out: 48px is the resolution where the budget binds most, with over half the runs peaking at the final epoch against 16–27% elsewhere. The accuracy still being won in the closing epochs there is however

Table S25: Convergence by target resolution. "Still improving" counts runs whose best validation accuracy occurred at the final or penultimate epoch.

Resolution	Runs	Mean best epoch	Still improving	Gain in last 3 epochs
48px	36	6.97	52.8%	1.38pp
96px	96	5.08	24.0%	4.21pp
112px	93	4.94	16.1%	3.89pp
160px	95	5.79	27.4%	3.27pp
224px	183	5.66	23.0%	2.89pp

the smallest of any resolution, 1.38pp, so the runs were converging slowly rather than being cut off mid-ascent. We report the 48px numbers with this limitation stated rather than claiming convergence.

S19 Sensitivity of the TDS Ranking to Its Definition

[New in this revision, in response to the reviews.]

TDS was criticised for using only the endpoints of the stride curve, for depending on baseline accuracy and floor effects, and for discarding negative drops through its positive part. Each objection defines a variant, and none changes the ranking (Table S26).

Table S26: Dataset demand under five definitions, and the rank correlation of each against the published TDS. *unclipped* removes $[\cdot]^+$; *AUC* integrates the whole stride curve trapezoidally in $\log_2 s$; *relative* divides by stride-1 accuracy; *normalised* divides by the headroom above chance.

Dataset	Published	Unclipped	AUC	Relative	Normalised
FineGym	55.92	55.92	24.33	76.92	78.03
AUTSL	53.02	53.02	17.74	71.22	71.65
SSv2	27.57	27.57	9.61	65.31	66.23
DriveAct	21.88	21.88	8.84	31.87	33.35
Diving-48	19.16	19.16	5.10	45.18	47.66
HMDB-51	16.64	16.64	4.94	21.05	21.58
EPIC-Kitchens	9.74	9.74	3.02	26.86	27.73
UCF-101	4.88	4.88	0.81	5.54	5.61
Spearman vs. published	—	1.000	1.000	0.952	0.952

The positive part never activates: no dataset has a negative mean drop, so *unclipped* is numerically identical to the published metric. Integrating the full curve preserves the ordering exactly. Normalising by baseline accuracy or by headroom above chance exchanges only adjacent pairs (DriveAct with Diving-48, HMDB-51 with EPIC-Kitchens); the three highest-demand datasets are identical under all five definitions.

S20 Multi-Stride Augmentation Ablation

[New in this revision, in response to the reviews.]

A reviewer asked whether the measured cliffs reflect a mismatch between the training and inference sampling distributions rather than anything intrinsic. We tested this directly by fine-tuning with random $(\mathcal{C}, s)$ exposure drawn from the same grid used at evaluation, and comparing against the published checkpoint. All three conditions are evaluated under the *published* sampler: evaluating the augmented arms without its padding would remove the cliff by construction and answer nothing.

Two augmented arms are trained. **paper** draws its augmentation through the published sampler, padding included; **fixed** draws from the same grid but resamples uniformly, with no repeated final frame. Within each architecture the two arms reach almost identical dense-sampling accuracy (TimeSformer 64.0% and 63.6%), so any difference in the stride curve is attributable to the form of the augmentation rather than to one arm having trained better.

Table S27: Both architectures on AUTSL, evaluated under the published sampler at coverage 100%. "Recovered" is the share of the baseline stride-1 to stride-16 cliff regained at stride 16; "dense cost" is the accuracy given up at stride 1.

Model	Arm	s=1	s=2	s=4	s=8	s=16	Cliff	Recov.	Dense cost
TimeSformer	baseline	66.8	66.9	66.3	65.7	50.8	16.0	—	—
	paper	63.5	63.6	63.1	62.6	55.3	8.2	28.4%	−3.3
	fixed	63.1	63.4	63.4	62.3	52.6	10.4	11.7%	−3.7
R3D-18	baseline	75.0	75.3	68.0	27.2	6.5	68.5	—	—
	paper	65.1	65.3	62.1	44.9	31.9	33.2	37.0%	−9.9
	fixed	63.8	63.2	59.9	28.0	10.4	53.3	5.7%	−11.2

On AUTSL for these two architectures, multi-stride exposure materially reduces the measured cliff: the paper-sampler arm recovers 28.4% for TimeSformer and 37.0% for R3D-18. This establishes a train–inference distribution effect, but it is not a causal decomposition of the remaining loss. The augmented models retain 63–72% of the baseline cliff and give up 3.3pp (TimeSformer) or 9.9pp (R3D-18) at stride 1, so the intervention trades dense and sparse operating points rather than producing universal robustness.

The comparison between the two augmented arms is consistent with sensitivity to sampler details. Both see the same nominal $(\mathcal{C}, s)$ grid and achieve similar dense accuracy, but the arm trained through the published repeat-last sampler recovers more at stride 16 than the fixed-length arm. This is consistent with adaptation to the train–inference input distribution. It does not, by itself, identify a spectral mechanism or prove that the residual loss is caused by missing temporal evidence. Broader model–dataset ablations would be needed for either conclusion.

We therefore retain the conclusion that the dominant term is evidence loss, with the correction that a recoverable mismatch term of 28–37% sits alongside it.

S21 Varying the Sampling Rate at Fixed Frame Count

[New in this revision, in response to the reviews.]

A reviewer observed that our temporal protocol investigates "how early the model has enough temporal context, rather than the aliasing due to frame sampling stride". This sweep tests that reading directly. The frame count k is pinned to each model's native budget and only the width W of a centred window varies, so the realised sampling rate k/W changes by an order of magnitude while the number of frames delivered never changes.

At fixed k, widening W lowers the sampling rate but also admits more of the action. These push in opposite directions, and no frame-subsampling design can separate them — that is the identifiability limitation stated in Sec. **??**. What the sweep does decide is which one binds empirically.

Table S28: Top-1 (%) against window width, averaged over architectures, with k fixed at each model's frame budget. Widening the window reduces the sampling rate tenfold, yet accuracy rises in seven of eight datasets.

Dataset	W=10%	25%	50%	75%	100%	$\rho(W, \text{acc})$	Δ
AUTSL	17.9	37.2	61.6	71.6	68.3	+1.000	+50.4
SSv2	8.3	17.7	29.5	36.1	38.4	+1.000	+30.1
Diving-48	12.8	22.3	38.3	43.2	41.5	+0.900	+28.7
HMDB-51	56.5	68.0	76.5	80.4	80.2	+0.900	+23.7
DriveAct	49.1	57.1	63.0	66.6	68.7	+1.000	+19.6
EPIC-Kitchens	17.4	24.8	31.5	35.2	36.4	+1.000	+19.0
UCF-101	77.9	85.0	88.9	90.0	89.9	+0.900	+12.0
Kinetics-400	61.5	64.5	64.6	64.1	63.1	+0.100	+1.6

Seven of eight datasets are context-bound, four of them at $\rho = +1.000$: giving a model the same number of frames spread over the whole action beats giving it the same number of frames densely covering a tenth of it, by as much as 56pp. If undersampling were the dominant cost, the sign would be the other way round. Over the range this protocol can reach, the binding constraint is how much of the action is observed, not how densely it is sampled — which is the reviewer's reading, and we adopt it.

Kinetics-400 is the single exception, and it is flat rather than reversed ($\rho = +0.100$, +1.6pp across the whole range). This is interpretable: its clips are fixed 10-second segments with the labelled action near the centre, so the narrowest window already contains the discriminative evidence and widening it mostly admits unrelated context. That one dataset sits at zero while the other seven rise steeply is evidence that the measure discriminates rather than merely trending positive.

We stress what this does *not* establish. Because W moves context and rate together, a density effect cannot be ruled out, only shown to be dominated. We therefore use this experiment to delimit the claim rather than to assert a literal spectral mechanism.

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